

Carbon Credit Auction Clearing via Probabilistic Computing

Quantum Dice Trinity Challenge 2026 Stage 3 Submission by Ballu Di Team

1. Problem, Data, and Formulation

The problem context, data source, bid reconstruction methodology, and Hamiltonian formulation are established in our Stage 2 submission, which we do not repeat here. In brief: 321 synthetic bids are reconstructed across four EU ETS credit types (EUA, EUA_DE, EUA_PL, EUAA) from EEX session aggregates using a Pareto quantity distribution and triangular price distribution. The clearing Hamiltonian is $H = H_{\text{obj}} + H_{\text{pkg}} + H_{\text{cap}}$, with welfare objective H_{obj} , package synergy bonus H_{pkg} for cross-type compliance pairs, and squared supply penalty H_{cap} with coefficient C .

Stage 3 introduces two refinements to the penalty structure and feasibility enforcement described below.

2. Refinements

Per-Type Penalty Coefficients

Stage 2 derived a single global penalty coefficient C from the spin-flip dominance condition:

$$C > \frac{A(v_{\max} + w_{\max})}{2 q_i n_{\text{acc}} \bar{q}} \quad (1)$$

evaluated at pooled statistics $\bar{q} = 0.022$ across all types. This is structurally insufficient when bid-quantity distributions are heterogeneous across credit types. Empirically:

Type	n	$\bar{q}$	$q_{\max}$	q_{sum}
EUA	100	0.0139	0.2859	1.39
EUA_DE	90	0.0242	0.6524	2.18
EUA_PL	82	0.0210	0.2296	1.72
EUAA	49	0.0337	0.3258	1.65

EUA_DE contains a single bid with $q_{\max} = 0.652$, consuming 65.2% of that type’s total supply in a single bid. This is a feature of real EU ETS markets: large industrial operators submit block bids covering their entire annual compliance obligation. A global C derived from the pooled mean $\bar{q} = 0.022$ provides penalty pressure approximately $0.652/0.022 \approx 30\times$ too weak for this bid specifically, as equation (1) scales inversely with q_i .

We therefore derive a per-type penalty coefficient C_τ for each credit type independently, applying equation (1) using each type’s own $v_{\max}^\tau$, $w_{\max}$, and $q_{\max}^\tau$ as the worst-case denominator:

$$C_\tau = C_{\text{mult}} \cdot \frac{v_{\max}^\tau + w_{\max} + 1}{2 q_{\max}^\tau} \quad (2)$$

Using $q_{\max}^\tau$ rather than $\bar{q}$ ensures the dominance condition holds for the single largest bid in each type, which is the binding constraint under a heavy-tailed distribution. The resulting from our toy example (present in the submitted notebook) coefficients are:

Type	C_τ
EUA	352.5
EUA_DE	154.5
EUA_PL	438.9
EUAA	309.3

The QUBO matrix is constructed with each type’s C_τ applied to its own capacity penalty block. H_{obj} and H_{pkg} are unchanged.

Classical Post-Processing Repair

Penalty-based QUBO encoding enforces supply constraints as soft penalties. For heavy-tailed bid distributions, even a well-calibrated C_τ admits occasional feasibility violations at the energy landscape boundary, where accumulated off-diagonal pushback from H_{cap} marginally fails to outweigh the objective reward for the largest bids.

Stage 2 addressed this with a 2% supply buffer ($S_\tau \rightarrow 0.98 S_\tau$ in the QUBO), targeting the real cap $S_\tau = 1.0$ while absorbing discretisation error. This is solver-specific calibration: the 2% margin was empirically tuned against `dwave-neal`’s annealing dynamics and has no principled basis for a different solver.

Stage 3 replaces the supply buffer with a solver-agnostic classical post-processing step. After the QUBO solver returns a candidate solution $\mathbf{x}$, feasibility is checked per type. Where violated, a greedy knapsack repair is applied: accepted bids in the infeasible type are ranked in descending order of value density v_i/q_i , and bids are removed from the bottom of this ranking until the constraint is satisfied:

$$\text{drop} = \arg \min_{i \in \mathcal{A}_\tau} \frac{v_i}{q_i}, \quad \text{until} \quad \sum_{i \in \mathcal{A}_\tau} q_i \leq S_\tau \quad (3)$$

This repair operator is independent of both the QUBO construction and the choice of solver, making the 2% buffer unnecessary. The welfare cost of repair is bounded by the value density of the dropped bids, which are by construction the least valuable per unit of supply consumed. The real supply cap $S_\tau = 1.0$ is restored in all feasibility checks.

3. Solvers

Three solvers are benchmarked against the same QUBO instance.

Integer Programming (IP): PuLP/CBC branch-and-bound. Solves the auction clearing problem exactly as a mixed-integer linear programme with native hard supply constraints, without QUBO encoding. Package bonus terms are linearised via auxiliary binary variables z_{ik} with standard McCormick constraints. Serves as the welfare optimality ceiling.

Simulated Annealing (SA): `dwave-neal` with 200 independent reads and 1000 sweeps per read. Operates on the QUBO formulation with per-type C_τ penalties and greedy repair post-processing.

Orbit Probabilistic Simulator: Quantum Dice `LocalSolver` implementing a CPU-based simulation of p-bit annealing [8]. Accepts the QUBO matrix directly as a `QuboProblem`. The matrix Q is normalised to $[-1, 1]$ by its maximum absolute entry before submission:

$$Q \leftarrow Q / \max_{ij} |Q_{ij}| \quad (4)$$

This prevents numerical overflow in the solver. Normalisation preserves the argmin of $E(\mathbf{x}) = \mathbf{x}^T Q \mathbf{x}$ and is there-

fore lossless with respect to solution quality. Solver parameters: `full_sweeps`= 1000, $\beta_{\text{start}} = 0.5$, $\beta_{\text{end}} = 12.0$, `beta_step_interval`= 10, `n_replicas`= 16.

Multi-Session Benchmark

Table 1 and Figure 1 report welfare and feasibility across 20 randomly sampled EUA sessions from the 2025 EEX dataset, each paired with fixed median EUA_DE, EUA_PL, and EUAA sessions. All three solvers return feasible solutions on every session after greedy repair post-processing.

SA achieves a mean welfare gap of 1.37% against IP across 19 sessions, with a range of 0.48%–2.81%, demonstrating consistent near-optimal performance across varying session statistics. Orbit achieves a mean welfare gap of 5.59% across the same 19 sessions, with a range of 3.66%–8.40%. One session (2025-11-18) is excluded from mean calculations: Orbit returned a 27.26% gap on this session, an outlier attributable to a deeply infeasible raw QUBO solution requiring aggressive greedy repair that cascaded into large welfare loss. This session is retained in Figure 1 for completeness.

Table 1: Welfare gap vs IP across 20 sessions (mean and range). Session 2025-11-18 excluded from Orbit statistics.

Solver	Mean gap (%)	Range (%)
SA (neal)	1.37	0.48 – 2.81
Orbit	5.59	3.66 – 8.40
IP	0.00	—

The persistent 4–5 percentage point gap between SA and Orbit across sessions is not attributable to a single problematic session or penalty miscalibration. A p-bit system annealed on a linear schedule freezes into the feasible region before the objective gradient has had sufficient time to differentiate high-welfare from low-welfare configurations. This is consistent with the result of Geman and Geman [12], who showed that convergence to the global optimum under Gibbs sampling requires an inverse-temperature schedule increasing no faster than $O(1/\log t)$, far slower than the linear schedule used here.

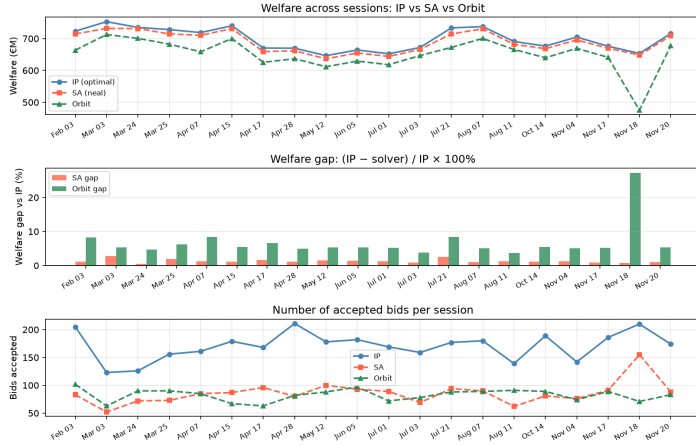


Figure 1: Welfare gap (%) vs IP across 20 sampled sessions. SA (neal) achieves a mean gap of 1.37%; Orbit achieves 5.59% (excluding the 27.26% outlier session on 2025-11-18).

Scaling Benchmark

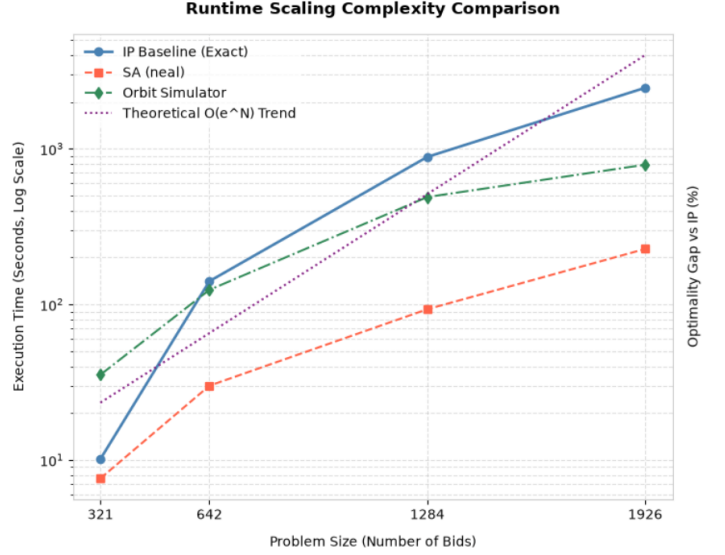


Figure 2: Runtime and welfare gap vs problem size n .

Scaling Benchmark

Figure 2 reports runtime and optimality gap across four problem sizes ($n \in \{321, 642, 1284, 1926\}$) obtained by scaling the base bid population by factors of $1\times$, $2\times$, $4\times$, and $6\times$.

Table 2: Runtime (seconds) and optimality gap (%) vs IP across problem sizes.

n	IP (s)	SA (s)	Orbit (s)	SA gap (%)
321	10.15	7.62	35.32	0.81
642	140.91	30.02	123.98	0.82
1284	886.50	93.09	489.52	1.13
1926	2465.40	227.17	790.00	0.64

Runtime: IP runtime grows super-linearly with problem size, consistent with the exponential worst-case complexity of branch-and-bound on binary programmes. SA runtime grows sub-linearly by comparison, from 7.62s to 227s over the same range a 29.8 $\times$ increase yielding a speedup over IP of 1.3 $\times$ at $n = 321$ growing to 10.8 $\times$ at $n = 1926$. Orbit runtime also grows more favourably than IP, reaching 790s at $n = 1926$ versus IP’s 2465s (3.1 $\times$ faster). This analysis is only for simulators and is not applicable for real hardware.

Solution quality: SA(neal) maintains a remarkably stable optimality gap of 0.64–1.13% across all four problem sizes, demonstrating that the QUBO penalty formulation with per-type C_r coefficients and greedy repair scales without quality degradation. Orbit’s gap improves with problem size, from 6.32% at $n = 321$ to 4.50% at $n = 1926$. This convergence behaviour is consistent with the relative penalty-to-objective ratio becoming more favourable as the number of bids per type grows: with more bids competing for the same supply, individual bid quantities q_i decrease on average. Whether Orbit’s gap continues to close toward SA at larger n remains an open question requiring access to physical probabilistic hardware beyond the scope of this submission.

4. Discussion

EUPHEMIA as a Future Target

The EU ETS primary auction is a uniform-price sealed-bid mechanism: the clearing decision reduces to sorting bids by price and accepting downward until supply is exhausted, an $O(N \log N)$ operation. The combinatorial structure introduced here via package bids is a modelling extension rather than an intrinsic property of the mechanism.

A structurally harder and more naturally motivated target is the EUPHEMIA algorithm, which clears the pan-European day-ahead electricity market across 25 countries. EUPHEMIA processes block bids all-or-nothing orders spanning contiguous delivery hours whose acceptance is conditional on the clearing prices of all hours in the block simultaneously. This creates mandatory pairwise couplings J_{ik} between block bids across periods that cannot be decomposed by hour or country, yielding a dense Ising Hamiltonian whose ground state encodes a globally consistent price-quantity equilibrium across the entire interconnected market.

We investigated implementing a EUPHEMIA block bid clearing instance using the ASSUME energy market simulation framework [11], specifically the `dam_with_complex_opt_clearing` configuration, recovering approximately 25 binary QUBO variables per day from block order data. The coupling structure between block acceptance, minimum acceptance ratios, and multi-period price consistency introduces non-quadratic interaction terms requiring auxiliary variable expansions via the Boros–Hammer method [4], expanding the variable count significantly beyond the base order count. Full EUPHEMIA-compatible QUBO construction is identified as the primary direction for future development.

5. Extensions Within Current Framework

Three extensions are identified within the existing QUBO structure. First, a 9-type model treating each EEX fund pool (IF, IX, MF, MX, SF) as a separate credit type τ with its own supply cap S_τ . Second, degree-3 package terms for three-way EUA–EUAA–EUA.DE compliance bundles, reducible to QUBO via the Boros–Hammer method [4] at one auxiliary variable per triple. Third, multi-session calibration across all 213 sessions in the 2025 dataset to test parameter stability across the observed price range of €70–84 per tCO₂.

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